Centered-encoding parametrization for the MMS sweep

Working note (not yet merged into `article.tex`)

2026-05-28

1 Motivation

The stabilized FE solver in this codebase is dimensional. Its inputs are $(\rho, \mu, \sigma, U, L, \boldsymbol{f})$ in a chosen system of units. The Method of Manufactured Solutions (MMS) test harness, however, organises runs by the *dimensionless* regime (Re, Da, α_0), and is free to choose any consistent dimensional encoding of each dimensionless cell. Here α_0 is the infimum of the porosity field $\alpha(x)$, and $\alpha_\infty = 1$ is its supremum (a convention shared with `article.tex`).

Until recently, the harness fixed

$$U = 1, \qquad L = 1, \tag{1}$$

and derived ν, σ from the dimensionless definitions used in `article.tex`:

$$\nu \;=\; \frac{U\,L}{Re} \;=\; \frac{1}{Re}, \qquad \sigma \;=\; \frac{Da\;\alpha_\infty\,\nu}{L^2} \;=\; \frac{Da\;\alpha_\infty}{Re}. \tag{2}$$

For the sweep grid $Re \in \{10^{-6}, 1, 10^6\}$, $Da \in \{10^{-6}, 1, 10^6\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$ (with $\alpha_\infty = 1$), this encoding produces values such as $\nu = 10^6$ and $\sigma = 10^{12}$ for the cell $(Re = 10^{-6}, Da = 10^6, \alpha_0 = 1)$. Twelve orders of magnitude separate the smallest from the largest dimensional coefficient that the solver sees, *all on the same side of unity.*

While the dimensionless physics is unchanged, this one-sided spread breaks numerical components of the stabilized solver that are not scale-invariant once round-off enters:

- the OSGS staggered fixed-point's L^2 projection π_h of the strong residual evaluates $\sigma\,\boldsymbol{u}_h$ and $\nu\Delta\boldsymbol{u}_h$ at near-precision-floor $\boldsymbol{u}_h$ when $\boldsymbol{u}_h \sim 1/\sigma$, amplifying floating-point noise into a non-contractive π_h update sequence;
- the assembled Jacobian's row magnitudes span 12 orders, which the direct factorisation absorbs but the Newton merit function and line-search tolerances do not;
- the MMS forcing oracle, computed by symbolic differentiation of the analytic exact solution, mixes $\sigma\,\boldsymbol{u}_{\mathrm{ex}}$ at magnitude 10^{12} with sub-dominant terms at 10^6, losing six digits of relative precision in the sub-dominant contributions.

2 The centered encoding

For a fixed dimensionless cell (Re, Da, α_∞), the harness has *two* free dimensional parameters (U, L). The freedom follows from the constraint pair (matching `article.tex` eq. 3.4):

$$Re = \frac{U\,L}{\nu}, \qquad\qquad Da = \frac{\sigma\,L^2}{\alpha_\infty\,\nu}. \tag{3}$$

Eliminating ν, σ:

$$\nu\,\sigma \;=\; \frac{\alpha_\infty\; Da\; U^2}{Re^2}, \qquad \frac{\nu}{\sigma} \;=\; \frac{L^2}{\alpha_\infty\; Da}. \tag{4}$$

So U controls the geometric mean and L controls the spread.

Choice. The *centered encoding* chooses U so that $\sqrt{\nu\,\sigma} = 1$, leaving $L = 1$ because the manufactured solution shape $\sin(\pi x)$ is hard-coded for a unit domain (see `Paper2DMMS` in `src/problems/mms_paper_2d.jl`):

$$\boxed{U \;=\; \frac{Re}{\sqrt{\alpha_\infty\; Da}}, \qquad L \;=\; 1.} \tag{5}$$

The resulting dimensional coefficients are

$$\nu \;=\; \frac{1}{\sqrt{\alpha_\infty\; Da}}, \qquad \sigma \;=\; \sqrt{\alpha_\infty\; Da}. \tag{6}$$

One checks immediately that $\nu\,\sigma = 1$ (centering condition) and $\nu/\sigma = 1/(\alpha_\infty\; Da)$ (intrinsic spread). Note that ν and σ no longer depend on Re — only U absorbs the Re variation. The spread $\sigma/\nu = \alpha_\infty\; Da$ is *intrinsic* to the cell and cannot be compressed by any choice of (U, L); centering only shifts it to be symmetric about unity.

What this does *not* change. The dimensionless equation (and therefore the dimensionless solution $\boldsymbol{u}^*, p^*$) is identical between the default and centered encodings; this is a strict reparametrization. The discrete velocity *degrees of freedom* are $\boldsymbol{u}_h^{\text{node}} \approx U\,\boldsymbol{u}^{*,\text{node}}$, so they take on different absolute magnitudes between encodings, but their dimensionless counterparts $\boldsymbol{u}^{*,\text{node}}$ are unchanged. The H^1-seminorm error $|\boldsymbol{u}_h - \boldsymbol{u}_{\text{ex}}|_{H^1(\Omega)}\,/\,(U/L)$ is regime-invariant (verified empirically: agreement to $<1\%$ across encodings on the easy regime C2).

3 Per-cell parameter table

The harness sweeps $Re \in \{10^{-6}, 1, 10^6\}$, $Da \in \{10^{-6}, 1, 10^6\}$, $\alpha_0 \in \{1, 0.5, 0.05\}$, with $\alpha_\infty = 1$ in all cells. Three cells are skipped by config ($(Re, Da, \alpha_0) = (10^6, \star, 0.05)$, three triples), leaving 24 that are exercised by the sweep. Table 1 reports the dimensional values for both encodings. Because the dimensional σ in the code path (`sigma_c = Da * alpha_infty * nu_calc / L^2`) uses $\alpha_\infty = 1$ throughout, the σ columns are independent of α_0 — they are determined purely by (Re, Da).

Maximum dimensional coefficient. The worst cells under the default encoding (C7, C16, C23) hit $\sigma = 10^{12}$. Under the centered encoding the worst is $\sigma = 10^3$ (any cell with $Da = 10^6$). The maximum dimensional coefficient drops by nine orders of magnitude across the sweep.

4 Verification evidence

Two single-cell probes (`test/extended/ManufacturedSolutions/diagnostics/jacobian_equilibration_osgs_` validated the reparametrization before the full rerun. Note that the probe used a slightly different centering formula $U = Re\,/\sqrt{\alpha_0\, Da}$ (with $\alpha_0 = 0.5$ for C16), so $U = 1.41 \cdot 10^{-9}$ and $\sigma = 1.41 \cdot 10^3$ in the probe; the harness rerun uses $U = Re\,/\sqrt{\alpha_\infty\, Da}$ with $\alpha_\infty = 1$, giving $U = 10^{-9}$ and $\sigma = 10^3$ exactly.

#	Re	Da	α_0	U	ν	σ	U	ν	σ
	Dimensionless inputs			Default ($U = L = 1$)			Centered		
C1	10^{-6}	10^{-6}	1.00	1	10^6	1	10^{-3}	10^3	10^{-3}
C2	1	10^{-6}	1.00	1	1	10^{-6}	10^3	10^3	10^{-3}
C3	10^6	10^{-6}	1.00	1	10^{-6}	10^{-12}	10^9	10^3	10^{-3}
C4	10^{-6}	1	1.00	1	10^6	10^6	10^{-6}	1	1
C5	1	1	1.00	1	1	1	1	1	1
C6	10^6	1	1.00	1	10^{-6}	10^{-6}	10^6	1	1
C7	10^{-6}	10^6	1.00	1	10^6	10^{12}	10^{-9}	10^{-3}	10^3
C8	1	10^6	1.00	1	1	10^6	10^{-3}	10^{-3}	10^3
C9	10^6	10^6	1.00	1	10^{-6}	1	10^3	10^{-3}	10^3
C10	10^{-6}	10^{-6}	0.50	1	10^6	1	10^{-3}	10^3	10^{-3}
C11	1	10^{-6}	0.50	1	1	10^{-6}	10^3	10^3	10^{-3}
C12	10^6	10^{-6}	0.50	1	10^{-6}	10^{-12}	10^9	10^3	10^{-3}
C13	10^{-6}	1	0.50	1	10^6	10^6	10^{-6}	1	1
C14	1	1	0.50	1	1	1	1	1	1
C15	10^6	1	0.50	1	10^{-6}	10^{-6}	10^6	1	1
C16	10^{-6}	10^6	0.50	1	10^6	10^{12}	10^{-9}	10^{-3}	10^3
C17	1	10^6	0.50	1	1	10^6	10^{-3}	10^{-3}	10^3
C18	10^6	10^6	0.50	1	10^{-6}	1	10^3	10^{-3}	10^3
C19	10^{-6}	10^{-6}	0.05	1	10^6	1	10^{-3}	10^3	10^{-3}
C20	1	10^{-6}	0.05	1	1	10^{-6}	10^3	10^3	10^{-3}
(skipped: $Re = 10^6, Da = 10^{-6}, \alpha_0 = 0.05$)									
C21	10^{-6}	1	0.05	1	10^6	10^6	10^{-6}	1	1
C22	1	1	0.05	1	1	1	1	1	1
(skipped: $Re = 10^6, Da = 1, \alpha_0 = 0.05$)									
C23	10^{-6}	10^6	0.05	1	10^6	10^{12}	10^{-9}	10^{-3}	10^3
C24	1	10^6	0.05	1	1	10^6	10^{-3}	10^{-3}	10^3
(skipped: $Re = 10^6, Da = 10^6, \alpha_0 = 0.05$)									

Table 1: Dimensional parameter values for each dimensionless cell in the `phase1_quad_k1` sweep, under the default encoding ($U = L = 1$) and the centered encoding ($U = Re\,/\sqrt{\alpha_\infty\, Da}$, $L = 1$), with $\alpha_\infty = 1$ as run by the test harness (`run_test.jl:76`). The dimensionless physics is identical between the two columns; only the dimensional representation differs. Under the centered encoding, ν and σ depend only on Da, never on Re or α_0.

C16 OSGS (high-σ, the canonical fold case).

	Default	Centered (probe)
ν	10^6	$1.41 \cdot 10^{-3}$
σ	10^{12}	$1.41 \cdot 10^3$
OSGS staggered iters	25 (budget)	0 (immediate)
π_u-drift	$7.6 \cdot 10^8$	$1.03 \cdot 10^{-6}$
`overall_converged`	`false`	`true`
Initial residual	$1.27 \cdot 10^{-3}$	$3.08 \cdot 10^{-10}$
$L^2(\boldsymbol{u})$	$1.07 \cdot 10^{-3}$	$4.72 \cdot 10^{-4}$
$H^1(\boldsymbol{u})$	$2.45 \cdot 10^{-1}$	$1.04 \cdot 10^{-1}$

The OSGS staggered fixed-point's projection drift drops by *fourteen orders of magnitude*. Same physics; the difference is the floating-point conditioning of the strong-residual projection.

C2 (easy regime; consistency check). With $Re = 1$, $Da = 10^{-6}$, $\alpha_0 = 1$, both encodings should give the same answer up to round-off.

	Default ($U = 1$)	Centered ($U = 10^3$)	
$H^1(\boldsymbol{u})$	$3.612 \cdot 10^{-2}$	$3.564 \cdot 10^{-2}$	(0.7% diff)
$L^2(\boldsymbol{u})$	$5.92 \cdot 10^{-4}$	$1.18 \cdot 10^{-4}$	(different noise floor)

The H^1 semi-norm error agrees to under 1% between encodings, confirming that the centered encoding represents the same discrete physics. The L^2 values differ by a constant noise-floor factor: in both runs the absolute residual at convergence is $O(10^{-9})$, and that absolute noise is divided by $U_c = U$ when forming the normalized error — a 1000× different denominator produces the apparent 5× discrepancy without any discretisation disagreement.

5 Implementation of centered encoding

The centered encoding alone is a single-cell-loop change in the MMS harness; no modification to the dimensional solver. The original `U_amp = 1.0`, `L = 1.0` line is replaced by:

```
# test/extended/ManufacturedSolutions/run_test.jl

(L_cell, U_amp) = compute_L_and_U(encoding_strategy,
                                  Re, Da, alpha_infty)
```

where `compute_L_and_U` returns $(L = 1, U = Re/\sqrt{\alpha_\infty Da})$ for the centered strategy (Section 2), $(L = 1, U = 1)$ for the legacy default, and the variable-L values for balanced and minmax (Section 6). The downstream pipeline (`build_mms_formulation`, `Paper2DMMS`, `evaluate_exactness_diagnostics`, `get_characteristic_scales`) threads U and L through correctly: the MMS exact velocity is multiplied by U, the manufactured forcing is computed analytically from that scaled solution, and error norms divide out the appropriate dimensionless combination (Section 6.9 corrects the naive divisor to give genuinely L-invariant quantities). The variable-L generalisation — polynomial frequency $\pi \to \pi/L$, domain bounding box $\Omega \to L \cdot \Omega$, porosity radii $r_i \to L \cdot r_i$, per-cell mesh rebuild — is the additional refactor documented in §6.6.

6 Variable L generalisation

The centered encoding of Section 2 spends the U degree of freedom and hardcodes $L = 1$ because the manufactured shape $\sin(\pi x)$ was wired to a unit period. Generalising the polynomial to $\sin(\pi x/L)$ and coupling the domain to L (so the dimensionless mesh ratio $h_{\text{dim}} = h/L = 1/N$ and the similarity analysis remain intact under any choice of L) unlocks the second degree of freedom.

Status (2026-06-01). The generalisation is implemented; the default `encoding_strategy = "centered"` preserves the pre-generalisation behaviour bit-identically, and the new strategies `"balanced"` and `"minmax"` expose the second-DOF machinery for use in MMS sweeps. The remainder of this section presents the math (§6.1–6.5), the implementation (§6.6), the empirical validation (§6.7), the Float64-precision interpretation (§6.8), and the dimensionless-error normalisation correction (§6.9).

6.1 Identities and the design space

Recall the two identities from Section 2:

$$\nu\,\sigma \;=\; \frac{\alpha_\infty\;Da\;U^2}{Re^2}, \qquad \frac{\nu}{\sigma} \;=\; \frac{L^2}{\alpha_\infty\;Da}. \tag{7}$$

The first identity says U alone controls the geometric mean $\sqrt{\nu\sigma}$ — L has no effect on it. The second says L alone controls the ratio ν/σ — U has no effect on it. The two degrees of freedom are therefore *orthogonal* in log-space: one slides the pair (ν, σ) along the line $\log\nu + \log\sigma = \text{const}$, the other rotates the pair around its geometric mean.

6.2 Balanced-coefficient encoding ($\nu = \sigma$)

The natural use of the L degree of freedom is to set $\nu/\sigma = 1$, making the diffusion and reaction coefficients dimensionally identical on every cell. From the second identity in (7):

$$L^\star \;=\; \sqrt{\alpha_\infty\;Da}. \tag{8}$$

Note that, like ν and σ in the centered encoding, $L^\star$ depends only on Da (and α_∞), not on Re or α_0.

The remaining choice is U. The centered selection $U = Re\,/\sqrt{\alpha_\infty\,Da}$ collapses the geometric mean to unity ($\sqrt{\nu\sigma} = 1$), so combined with (8) it yields $\nu = \sigma = 1$ on every cell. This is the most aggressive flattening of the operator coefficients, but it leaves U unchanged from the centered encoding — the same factor of 10^9 excursion to $U = 10^{-9}$ at the C7 corner.

6.3 Minmax encoding: trading U headroom against ν, σ

A different use of the L DOF is to minimise the worst-case excursion across all three of U, ν, σ simultaneously. Concretely, we solve

$$\min_{U,L}\;\; \max\!\big(\,|\log_{10} U|,\; |\log_{10}\nu|,\; |\log_{10}\sigma|\,\big). \tag{9}$$

Keeping $\nu = \sigma$ (which sets $L = L^\star$ as above) and choosing U to equalise $|\log U|$ with $|\log\nu|$ gives a closed form. Define

$$A \;\equiv\; \tfrac{1}{2}\log_{10}(\alpha_\infty\,Da) \;-\; \log_{10} Re\,. \tag{10}$$

A is the excursion of $\log \nu$ from $\log U$ at the centered choice. The minmax optimum is achieved at

$$\boxed{\log_{10} U^{\star\star} \;=\; -\frac{A}{2}, \qquad \log_{10} L^{\star\star} \;=\; \tfrac{1}{2}\log_{10}(\alpha_\infty\, Da), \qquad \log_{10} \nu^{\star\star} \;=\; \log_{10} \sigma^{\star\star} \;=\; +\frac{A}{2}.} \tag{11}$$

At this point, $|\log U| = |\log \nu| = |\log \sigma| = |A|/2$ — all three dimensional quantities sit at the same distance from unity, with U on the opposite side of σ in log-space. Closed forms:

$$U^{\star\star} \;=\; \left(\frac{Re^2}{\alpha_\infty\; Da}\right)^{1/4}, \qquad L^{\star\star} \;=\; \sqrt{\alpha_\infty\; Da}, \qquad \nu^{\star\star} \;=\; \sigma^{\star\star} \;=\; \left(\frac{\alpha_\infty\; Da}{Re^2}\right)^{1/4}. \tag{12}$$

Notice the satisfying symmetry $U^{\star\star} \cdot \nu^{\star\star} = 1$ on every cell.

Spread reduction. Under the centered encoding the worst-case excursion is $|A|$: $\log \nu = -A/2 \cdot 2 =$ (signed) A and $\log U =$ opposite-sign $A/2$, so $\max(|\log U|, |\log \nu|, |\log \sigma|) = |A|$. Under minmax that worst case drops to $|A|/2$ — **a factor-of-two reduction in log-magnitude**, i.e. the worst dimensional quantity halves its number of digits away from unity, on every cell.

6.4 Per-cell numerical values

Table 2 reports the dimensional values under both new strategies, alongside the centered baseline for reference. All values shown for $\alpha_\infty = 1$ as run by the harness.

	Dim'less			Centered ($L=1$)			Balanced ($\nu=\sigma$, U centered)				Minmax ($\nu=\sigma=1/U$)			
#	Re	Da	α_0	U	ν	σ	U	L	ν	σ	U	L	ν	σ
C1	10^{-6}	10^{-6}	1.00	10^{-3}	10^{3}	10^{-3}	10^{-3}	10^{-3}	1	1	$10^{-1.5}$	10^{-3}	$10^{1.5}$	$10^{1.5}$
C2	1	10^{-6}	1.00	10^{3}	10^{3}	10^{-3}	10^{3}	10^{-3}	1	1	$10^{1.5}$	10^{-3}	$10^{-1.5}$	$10^{-1.5}$
C3	10^{6}	10^{-6}	1.00	10^{9}	10^{3}	10^{-3}	10^{9}	10^{-3}	1	1	$10^{4.5}$	10^{-3}	$10^{-4.5}$	$10^{-4.5}$
C4	10^{-6}	1	1.00	10^{-6}	1	1	10^{-6}	1	1	1	10^{-3}	1	10^{3}	10^{3}
C5	1	1	1.00	1	1	1	1	1	1	1	1	1	1	1
C6	10^{6}	1	1.00	10^{6}	1	1	10^{6}	1	1	1	10^{3}	1	10^{-3}	10^{-3}
C7	10^{-6}	10^{6}	1.00	10^{-9}	10^{-3}	10^{3}	10^{-9}	10^{3}	1	1	$10^{-4.5}$	10^{3}	$10^{4.5}$	$10^{4.5}$
C8	1	10^{6}	1.00	10^{-3}	10^{-3}	10^{3}	10^{-3}	10^{3}	1	1	$10^{-1.5}$	10^{3}	$10^{1.5}$	$10^{1.5}$
C9	10^{6}	10^{6}	1.00	10^{3}	10^{-3}	10^{3}	10^{3}	10^{3}	1	1	$10^{1.5}$	10^{3}	$10^{-1.5}$	$10^{-1.5}$
$\alpha_0 = 0.5$ *cells (C10–C18): identical* U, ν, σ, L *values as C1–C9 because the harness uses* $\alpha_\infty = 1$.														
C19	10^{-6}	10^{-6}	0.05	10^{-3}	10^{3}	10^{-3}	10^{-3}	10^{-3}	1	1	$10^{-1.5}$	10^{-3}	$10^{1.5}$	$10^{1.5}$
C20	1	10^{-6}	0.05	10^{3}	10^{3}	10^{-3}	10^{3}	10^{-3}	1	1	$10^{1.5}$	10^{-3}	$10^{-1.5}$	$10^{-1.5}$
C21	10^{-6}	1	0.05	10^{-6}	1	1	10^{-6}	1	1	1	10^{-3}	1	10^{3}	10^{3}
C22	1	1	0.05	1	1	1	1	1	1	1	1	1	1	1
C23	10^{-6}	10^{6}	0.05	10^{-9}	10^{-3}	10^{3}	10^{-9}	10^{3}	1	1	$10^{-4.5}$	10^{3}	$10^{4.5}$	$10^{4.5}$
C24	1	10^{6}	0.05	10^{-3}	10^{-3}	10^{3}	10^{-3}	10^{3}	1	1	$10^{-1.5}$	10^{3}	$10^{1.5}$	$10^{1.5}$

Table 2: Per-cell dimensional values under three encodings: centered ($L = 1$ fixed), balanced ($\nu = \sigma$ via $L = \sqrt{\alpha_\infty\, Da}$ with centered U), and minmax (jointly optimal U and L). Skipped cells ($Re = 10^6, \alpha_0 = 0.05$) omitted. Values shown with $\alpha_\infty = 1$ as run by the test harness; under this convention all three encodings are independent of α_0.

6.5 Comparison summary

For each encoding and each cell, the worst-case dimensional excursion $M = \max(|\log_{10} U|, |\log_{10} \nu|, |\log_{10} \sigma|)$ measures how far from unity the most extreme coefficient lies. The sweep maxima over all 24 cells are:

Encoding	$\max_{\text{cells}} M$	Worst-case cells
Default ($U = L = 1$)	12	C3, C7, C12, C16, C23
Centered ($L = 1$)	9	C3, C7, C12, C16, C23
Balanced ($\nu = \sigma$, U centered)	9	C3, C7, C12, C16, C23 (set by $\lvert \log U \rvert$)
Minmax (jointly optimal U, L)	4.5	C3, C7, C12, C16, C23

The same five $(\mathrm{Re}, \mathrm{Da}) \in \{(10^6, 10^{-6}), (10^{-6}, 10^6)\}$ corner cells dominate in every encoding — they are the diagonally-opposite corners of the $(\mathrm{Re}, \mathrm{Da})$ plane that the sweep grid samples. The balanced encoding flattens the operator coefficients to $\nu = \sigma = 1$ but does not improve the worst-case excursion because $\lvert \log U \rvert$ is unchanged from centered. The minmax encoding redistributes the excursion across all three quantities and halves the worst case from 9 to 4.5 orders of magnitude.

6.6 Implementation

The generalisation was implemented on 2026-06-01. The diff is contained to two files plus blitz tests:

`src/problems/mms_paper_2d.jl`. The hard-coded spatial frequency π is replaced by $k = \pi/L$ throughout the analytic chain. The struct `UExFunc` gains an `L::Float64` field; the operator $f \mapsto f(x) = U\alpha_0\alpha(x)^{-1}S(x/L)$ is now genuinely L-aware. The five analytic-derivative functions `grad_u_ex`, `lap_u_ex`, `grad_div_u_ex`, `get_p_ex`, and the pressure-gradient block in `evaluate_exactness_diagnosti` each pick up the appropriate $1/L$ from the chain rule: first derivatives by $1/L$, second derivatives by $1/L^2$. A backward-compatible three-argument constructor `UExFunc(U, alpha_0, alpha_field)` with $L = 1$ default is added so legacy probes that pre-date the generalisation continue to compile.

`test/extended/ManufacturedSolutions/run_test.jl`. A new helper `compute_L_and_U(strategy, Re, Da, alpha_infty)` returns the pair (L, U_{amp}) for the four named strategies `"centered"` (default — $L = 1$, $U = \mathrm{Re}/\sqrt{\alpha_\infty \mathrm{Da}}$), `"balanced"` ($L = \sqrt{\alpha_\infty \mathrm{Da}}$, same U), `"minmax"` ($L = \sqrt{\alpha_\infty \mathrm{Da}}$, $U = (\mathrm{Re}^2/(\alpha_\infty \mathrm{Da}))^{1/4}$), and `"unit"` ($L = U = 1$, kept for diagnostic probes). The strategy is read once at the top of `run_mms` via `get(test_dict, "encoding_strategy", "centered")`.

The harness builds the per-cell L *inside* the $(\mathrm{Re}, \mathrm{Da}, \alpha_0)$ triple loop because L varies per cell under non-centered strategies. The mesh and FE spaces are therefore rebuilt per cell rather than once per N — a few extra seconds of wall-clock per sweep, negligible against solve time. `_build_local_mesh` accepts L and tuples `L .* domain_cfg.bounding_box`; `build_porosity_field` accepts L and constructs `SmoothRadialPorosity` with $L \cdot r_1$ and $L \cdot r_2$. The homotopy perturbation `h_raw_func` is rescaled with $k = \pi/L$ on the polynomial frequencies and with $1/L^8$ on the boundary bubble B_{fn}, so $\|h\|_{L^2(\Omega)}$ stays $O(L^{d/2})$ matching the manufactured solution.

Similarity invariants preserved by the implementation. The key check that the dimensionless physics is genuinely the same across encodings:

- $\mathrm{Re}_{\text{check}} = U \cdot L/\nu = UL/(UL/\mathrm{Re}) = \mathrm{Re}$ — preserved algebraically.
- $\mathrm{Da}_{\text{check}} = \alpha_\infty \sigma L^2/\nu = \alpha_\infty(\alpha_\infty \mathrm{Da}\nu/L^2)L^2/(\alpha_\infty \nu) = \mathrm{Da}$ — preserved.
- $u_{ex}(L\hat{x})/U_{\text{amp}} = \alpha_0/\alpha(L\hat{x}) \cdot \hat{S}(\hat{x})$. For this ratio to be L-invariant, $\alpha(L\hat{x})$ as a function of $\hat{x}$ must equal $\alpha(\hat{x})$ at $L = 1$. Writing $\tilde{r}_i = L \cdot r_i$ for the L-scaled radii stored in the field (set by

`build_porosity_field`), the smoothing parameter at physical r is $\eta(r;L) = (r^2 - \tilde{r}_1^2)/(\tilde{r}_2^2 - \tilde{r}_1^2)$. At $r = L\hat{r}$:

$$\eta(L\hat{r};L) \;=\; \frac{L^2\hat{r}^2 - L^2 r_1^2}{L^2 r_2^2 - L^2 r_1^2} \;=\; \frac{\hat{r}^2 - r_1^2}{r_2^2 - r_1^2} \;=\; \eta(\hat{r};1).$$

The L^2 factor cancels between numerator and denominator, so α at the corresponding dimensionless position is identical to the $L = 1$ baseline.

- Physical $\nabla_x \alpha$ scales as $1/L$ — exactly as a physical gradient should under coordinate dilation. From the same formula, $d\eta/dr = 2r/(\tilde{r}_2^2 - \tilde{r}_1^2) = 2r/(L^2(r_2^2 - r_1^2))$. At $r = L\hat{r}$ this gives $(2\hat{r}/(r_2^2 - r_1^2)) \cdot 1/L$, so the dimensionless $L \cdot \nabla\alpha = 2\hat{r}/(r_2^2 - r_1^2)$ is encoding-invariant.
- The grid Reynolds number $\mathrm{Re}_h = Uh/\nu = (UL/\nu) \cdot (h/L) = \mathrm{Re}/N$ — independent of L. Stabilisation τ_{NS} at the same N should therefore have the same dimensionless value across encodings.

6.7 Empirical validation

Two end-to-end checks were performed.

$L = 1$ bit-identical regression. Cell C5 (Re$=1$, Da$=1$, $\alpha_0 = 1$, OSGS) run at $N \in \{10, 20, 40, 80\}$ through the new harness with `encoding_strategy = "centered"` reproduces the master HDF5 values to all seven reported digits:

	Master			New harness (centered)		
N	L^2_u	L^2_p	H^1_u	L^2_u	L^2_p	H^1_u
10	9.530767×10^{-3}	2.298378×10^{-2}	2.858131×10^{-1}	9.530767×10^{-3}	2.298378×10^{-2}	2.858131×10^{-1}
20	1.841275×10^{-3}	1.928104×10^{-2}	1.438814×10^{-1}	1.841275×10^{-3}	1.928104×10^{-2}	1.438814×10^{-1}
40	4.633695×10^{-4}	5.695171×10^{-3}	7.150274×10^{-2}	4.633695×10^{-4}	5.695171×10^{-3}	7.150274×10^{-2}
80	1.313346×10^{-4}	1.976955×10^{-3}	3.568200×10^{-2}	1.313346×10^{-4}	1.976955×10^{-3}	3.568200×10^{-2}

The agreement confirms that the per-cell mesh rebuild, the $k = \pi/L$ substitution, the porosity-radius scaling, and the normalisation correction (§6.9) collapse exactly to the pre-generalisation formula at $L = 1$.

C7 minmax probe. The corner cell C7 (Re$=10^{-6}$, Da$=10^6$, $\alpha_0 = 1$, OSGS) was the empirical motivation: under the centered encoding, the L^2_u slope at the finest mesh pair drops from 2.04 at $N = 80 \to 160$ to 0.84 at $N = 160 \to 320$ — a stall attributed in Section 4 to the solution-scale precision floor at $U_{\mathrm{amp}} = 10^{-9}$. Running the same cell at `encoding_strategy = "minmax"` (which gives $U_{\mathrm{amp}} = 10^{-4.5}$, $L = 10^3$, $\nu = \sigma = 10^{4.5}$) yields the complementary trajectory:

$N_{\mathrm{coarse}} \to N_{\mathrm{fine}}$	Centered slope	Minmax slope
$10 \to 20$	1.96	0.94
$20 \to 40$	1.98	1.28
$40 \to 80$	2.01	1.47
$80 \to 160$	2.04	1.63
$160 \to 320$	**0.84** *(stalled)*	**1.93** *(recovered)*

Centered is clean at coarse meshes and stalls at the finest pair; minmax is pre-asymptotic at coarse meshes and recovers at the finest pair. The trajectories cross. The continuous problems and the dimensionless meshes are identical between the two; the difference is the Float64 realisation of the discrete operator, discussed next.

6.8 Floating-point realisation

The continuous dimensionless problem

$$-\tfrac{1}{Re}\widehat{\Delta}\hat{u} + \tfrac{\alpha_\infty\, Da}{Re}\hat{u} + (\hat{u}\cdot\widehat{\nabla})\hat{u} + \widehat{\nabla}\hat{p} = \hat{f} \tag{13}$$

is invariant under the choice of (U_{amp}, L) — both encodings discretise the same equation, and the algebraic similarity invariants of §6.6 (Re, Da, α at corresponding positions, $\nabla\alpha$ scaling, grid Reynolds $\mathrm{Re}_h = \mathrm{Re}/N$) all hold exactly. The discrete realisation is invariant only to the precision of the floating-point system. Float64's $\sim$ 16 digits of relative precision are preserved at each elementary operation, but the *same* dimensionless quantity computed via *different sequences of dimensional operations* at different orders of magnitude accumulates round-off differently. The empirical L=1 bit-identity (§6.7) and the dimensional-similarity algebra together pin the source to discrete operations whose absolute operand magnitudes change with the encoding; three plausible sites:

1. **OSGS strong-residual projection.** The discrete projection $\boldsymbol{\pi}_h = M^{-1}R(u_h)$ inverts the dimensional mass matrix M against the dimensional strong residual R. For C7 at $N = 10$, where the cell measure is $|\Omega|/N^2$:

   - Centered: $M_{ij} \sim |\Omega|/N^2 \sim 10^{-2}$; the residual entries R_i sum contributions $\sigma u_h \sim 10^{-6}$, $-\nu\Delta u_h \sim 10^{-11}$, $\nabla p_h \sim 3\cdot 10^{-6}$, all dimensional. The R_i extracted by quadrature is on order 10^{-8}–10^{-7}; the projected π on order 10^{-5}.
   - Minmax: $M_{ij} \sim L^2|\Omega|/N^2 \sim 10^4$; R_i summands are $\sigma u_h \sim 1$, $-\nu\Delta u_h \sim 10^{-5.5}$, $\nabla p_h \sim 0.3$ — five orders of magnitude apart in the σu vs $\nu\Delta u$ summands rather than centered's six.

   Both encodings produce the same dimensionless $\hat{\boldsymbol{\pi}}_h$ *exactly* (the dimensional rescaling cancels term by term), but the round-off in computing the dimensional R as a sum of four contributions at different magnitudes accumulates differently. The relative position of summands in log-space changes between encodings, and Float64 cancels the smaller ones differently when summed with the larger ones.

2. **Stabilisation parameter $\tau_{\mathbf{NS}}$.** The dimensional formula $\tau_{\mathrm{NS}} = (4c_1\nu/h^2 + 2c_2\sigma + c|u|/h)^{-1}$ contains summands that scale unequally per encoding. For C7 at $N = 10$:

   - Centered: $4c_1\nu/h^2 \approx 0.4c_1$, $2c_2\sigma \approx 2000c_2$ — summands span four orders.
   - Minmax: $4c_1\nu/h^2 \approx 12.6c_1$, $2c_2\sigma \approx 6.3\times 10^4 c_2$ — summands span four orders again, but the diffusion summand sits at a different relative weight and is more vulnerable to floating-point obliteration when added to the reaction summand.

   $\hat{\tau} = \tau \cdot U_c/L$ is encoding-invariant algebraically; the dimensional summation is not invariant in finite precision.

3. **Newton residual.** $\boldsymbol{r} = \boldsymbol{b} - \mathcal{J}\,\delta\boldsymbol{x}$ in centered has $|\boldsymbol{b}|, |\mathcal{J}\delta\boldsymbol{x}| \sim 10^{-7}$ cancelling to extract $|\boldsymbol{r}|$ at machine precision; in minmax the same cancellation happens between operands at $\sim 10^{4.5}$. Same relative residual reduction, different absolute round-off in the extraction.

At fine N the FE truncation grows back above these floating-point floors and the dimensionless slopes converge to the h^{k+1} optimum in *both* encodings. At coarse N and at the mesh where the corresponding floor is binding, encoding-dependent deviations show: centered fails first at solution-scale (C7 at $N = 320$ where $U_{\mathrm{amp}} h^{k+1} \sim \varepsilon_{\mathrm{mach}}$); minmax fails first at operator-scale (the same cell at coarse N where $\sigma h^2 \sim$ summand-cancellation noise). The full sweep of the 24 cells shows the bulk of cells lying away from either floor for both encodings; the slope trajectory is essentially identical for all cells where neither floor is binding (most of the $|\log\nu|, |\log\sigma|, |\log U| < 6$ interior).

6.9 Dimensionless error normalisation

The harness's error norms were historically computed as

$$\widetilde{e}_{L^2,u} \;=\; \frac{\|\boldsymbol{e}_u\|_{L^2(\Omega)}}{U_c}, \qquad \widetilde{e}_{H^1,u} \;=\; \frac{\|\nabla\boldsymbol{e}_u\|_{L^2(\Omega)}}{U_c/L}. \tag{14}$$

At $L = 1$ with a unit-area baseline bounding box these are numerically the dimensionless quantities $\|\hat{e}_u\|_{L^2(\hat{\Omega})}$ and $\|\widehat{\nabla}\hat{e}_u\|_{L^2(\hat{\Omega})}$, but at $L \neq 1$ they carry a residual factor of L — they are *not* fully dimensionless. Reading the L^2 similarity in detail ($\boldsymbol{e}_u(x) = U_c \cdot \hat{\boldsymbol{e}}_u(x/L)$, $\mathrm{d}x = L^d \mathrm{d}\hat{x}$):

$$\begin{aligned} \|\boldsymbol{e}_u\|^2_{L^2(\Omega)} \;&=\; U_c^2 \cdot L^d \cdot \|\hat{\boldsymbol{e}}_u\|^2_{L^2(\hat{\Omega})} \\ \Rightarrow \quad \|\boldsymbol{e}_u\|_{L^2(\Omega)} \;&=\; U_c \cdot \sqrt{|\Omega|} \cdot \|\hat{\boldsymbol{e}}_u\|_{L^2(\hat{\Omega})}. \end{aligned} \tag{15}$$

For the gradient, $\nabla_x \boldsymbol{e}_u = (U_c/L)\nabla_{\hat{x}}\hat{\boldsymbol{e}}_u$, and the L^2 similarity inherits a $(1/L)^2 \cdot L^d$ factor that in 2D reduces to unity:

$$\|\nabla\boldsymbol{e}_u\|_{L^2(\Omega)} \;=\; U_c \cdot \|\widehat{\nabla}\hat{\boldsymbol{e}}_u\|_{L^2(\hat{\Omega})} \qquad \text{(2D)}. \tag{16}$$

The dimensionless quantities are therefore

$$\boxed{\; e_{L^2,u} \;=\; \frac{\|\boldsymbol{e}_u\|_{L^2(\Omega)}}{U_c \cdot \sqrt{|\Omega|}}, \qquad e_{H^1,u} \;=\; \frac{\|\nabla\boldsymbol{e}_u\|_{L^2(\Omega)}}{U_c} \;} \tag{17}$$

(and analogously $P_c \cdot \sqrt{|\Omega|}$ and P_c for the pressure). The corrected formulas are implemented in `calculate_normalized_errors` in `test/extended/ManufacturedSolutions/run_test.jl` (line 150 of the 2026-06-01 commit). For the standard baseline domain $[-0.5, 0.5]^2$ at $L = 1$, $|\Omega| = 1$ and the corrected formulas collapse to the legacy ones bit-identically — which is exactly what the regression of §6.7 confirms.

7 What is deliberately *not* addressed here

- **Higher-precision arithmetic** as a Float64-floor remedy. The encoding-dependent slope drift documented in §6.8 is a property of Float64's 16-digit relative precision applied to dimensional operands at heterogeneous magnitudes. Switching the LU back-substitution (and the τ and OSGS-projection arithmetic) to extended precision — `DoubleFloats.jl` ($\sim$ 32 digits) or quad precision — would eliminate the floor at the cost of a $\sim$ 5–10× runtime overhead. This is a single-cell diagnostic worth running if a future paper requires the C7-corner slope cleanly at the finest mesh under *either* encoding; the variable-L machinery already moves the floor by 4.5 orders of magnitude, which is sufficient for the present sweep.

- **The pressure penalty $\eta = \text{eps\_val}$.** The dimensional pressure amplitude P_c varies with the encoding, which means $\eta \cdot p^2$ has a different effective weight per cell. Empirically the discrete LBB stabilization is the primary gauge fix and η acts only as a safety net; the rerun will validate this in practice. If the sweep reveals an η-related regression, a regime-aware scaling $\eta_{\text{cell}} \propto P_c^{-2}$ is a focused follow-up.

- **Solver tolerances.** `ftol`, `xtol`, `stagnation_noise_floor` are absolute. Under the centered encoding the residual magnitudes shrink for low-Re cells; the existing `dynamic_ftol_spatial_safety_factor` machinery scales `ftol` with h^{k+1}, which empirically reaches sensible values in both encodings (the probes converged to high precision). If a regression appears, the dynamic-tolerance formula is the first place to look.

Status (2026-06-01)

Centered encoding (Section 2) is the harness default and the `phase1_quad_k1` master HDF5 was produced under it. Variable-L generalisation (Section 6) is implemented in the harness with $L = 1$ centered preserved bit-identically; `"balanced"` and `"minmax"` are available as opt-in strategies. The corner case C7 (Re$= 10^{-6}$, Da$= 10^6$, $\alpha_0 = 1$) was the empirical diagnosis target: its Float64 noise-floor stall under centered at $N = 320$ is recovered by minmax at the same N, confirming that the mechanism is solution-scale precision (not algorithm-side). The remainder of the (Re, Da, α) sweep behaves equivalently across encodings for cells where neither floor is binding.

This document is kept in `theory/` as a standalone working note. Once a final paper-grade $k = 2$ sweep is performed (deciding between centered and minmax based on which side's Float64 floor is binding for that polynomial order), the contents here should be condensed into `article.tex` as a verification-section remark.